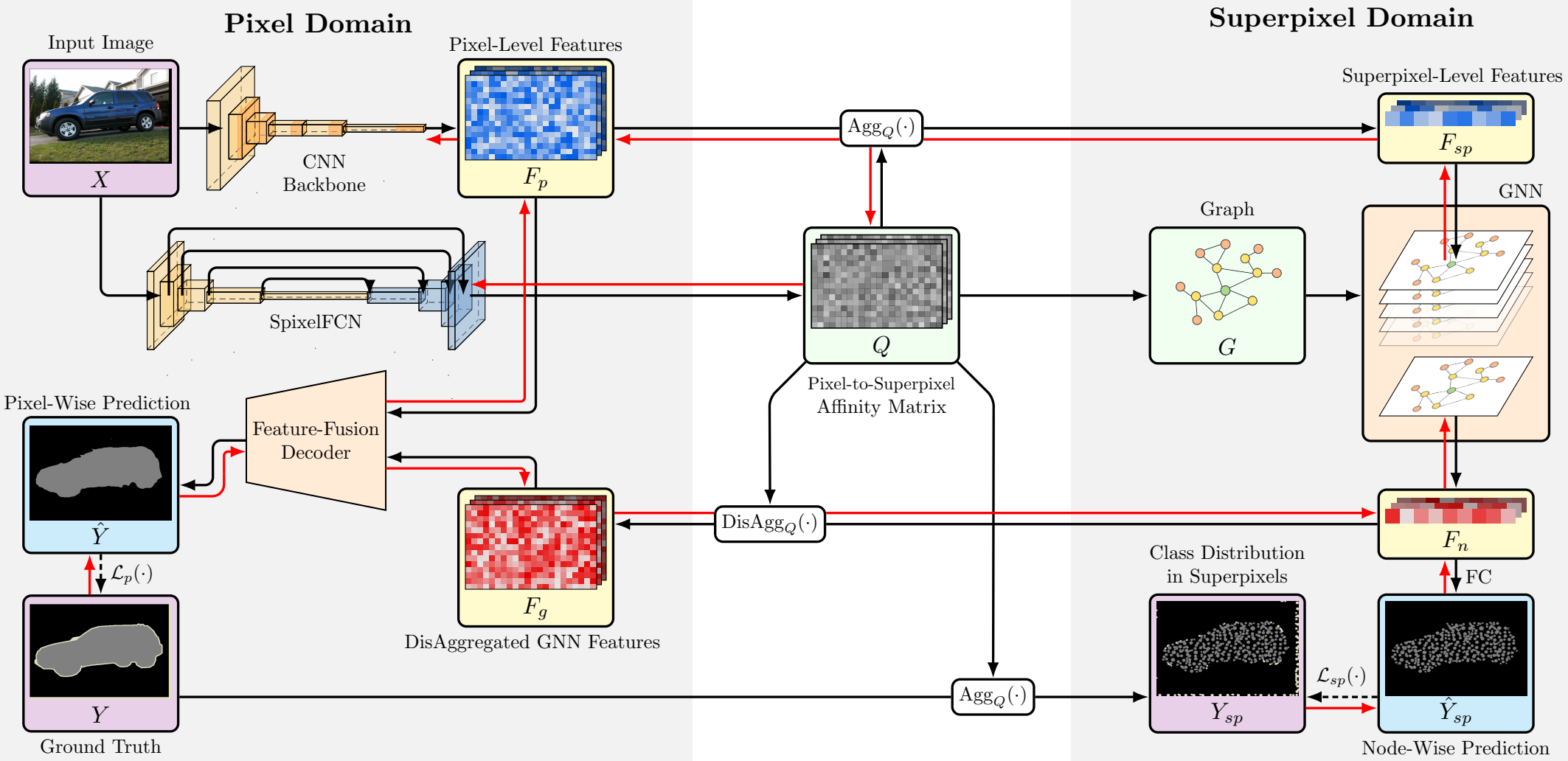
Pixel Domain
Superpixel Domain
Input Image
X
CNN Backbone
Pixel-Level Features
F_p
$\mathrm{Agg}_Q(\cdot)$
Superpixel-Level Features
F_{sp}
GNN
Graph
G
SpixelFCN
Q
Pixel-to-Superpixel Affinity Matrix
Pixel-Wise Prediction
Feature-Fusion Decoder
$\hat{Y}$
$\mathcal{L}_p(\cdot)$
$\mathrm{DisAgg}_Q(\cdot)$
F_n
FC
F_g
DisAggregated GNN Features
Class Distribution in Superpixels
Y
Ground Truth
$\mathrm{Agg}_Q(\cdot)$
$\mathcal{L}_{sp}(\cdot)$
Y_{sp}
$\hat{Y}_{sp}$
Node-Wise Prediction